

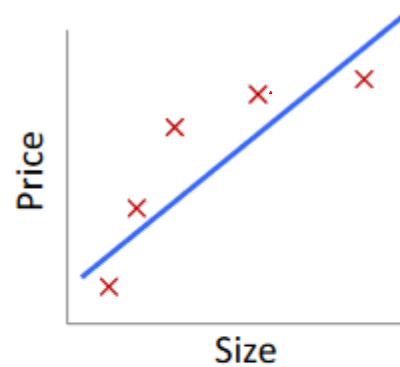
# Fitting Quality

$$h_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$

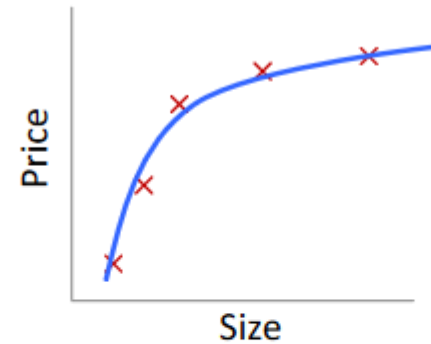
$$h_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3$$

- Underfitting: The learned hypothesis may not fit well (large  $J(\theta)$ )
- Overfitting: The learned hypothesis may fit well ( $J(\theta)$  very small)
  - Fails to generalize new example
  - Lots of feature and small amount of data

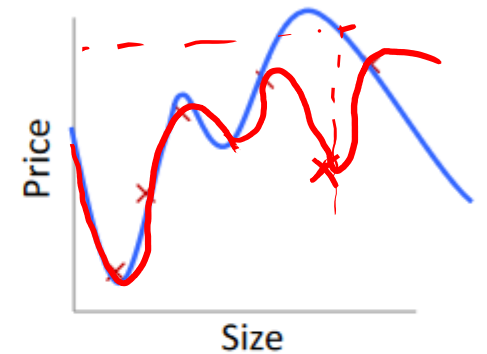
$x_1$  = Size of the house  
 $x_2$  = no. of bedrooms  
 $x_3$  = floor no.  
 $x_4$  = age of the house  
 $x_5$  = kitchen size  
 $x_6$  = . . .



$\theta_0 + \theta_1 x$   
 Underfitting  
 (high bias)



$\theta_0 + \theta_1 x + \theta_2 x^2$   
 Correct fit



$\theta_0 + \theta_1 x + \theta_2 x^2 + \theta_3 x^3 + \theta_4 x^4$   
 Overfitting  
 (high variance)

# Addressing Overfitting

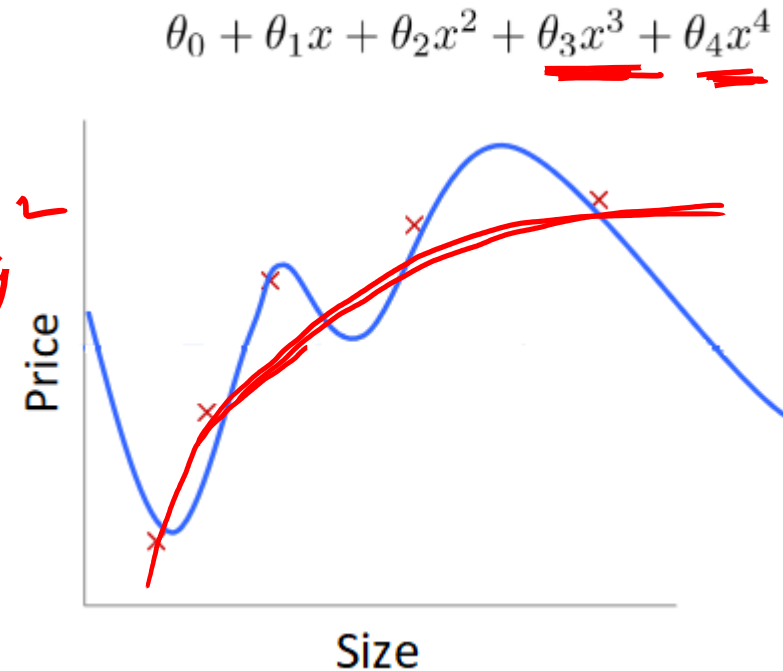
- Reduce no. of features  
(manually/model selection)

$$\min_{\theta} \frac{1}{2n} \sum_{i=1}^n \left( h_{\theta}(x^{(i)}) - y^{(i)} \right)^2 + \lambda_3 \theta_3^2 + \lambda_4 \theta_4^2$$

$$\lambda_3 = 1000$$

$$\lambda_4 = 1000$$

$$h_{\theta}(x) = \theta_0 + \theta_1 x + \theta_2 x^2$$



# Addressing Overfitting via Regularization

- A method for automatically controlling the complexity of the learned hypothesis
- Can be incorporated into the cost function
- Keeps all the features , but reduce magnitude/values of the parameters
- Works well when we have a lot of features, each that contributes a bit to predict the label
- Small values for parameters
  - Simpler hypothesis
  - Less-prone to overfitting

# Regularized Linear Regression (Ridge Regression) Objective Function

$$J(\boldsymbol{\theta}) = \underbrace{\frac{1}{2n} \sum_{i=1}^n (h_{\boldsymbol{\theta}}(\mathbf{x}^{(i)}) - y^{(i)})^2}_{\text{"Data"}} + \underbrace{\frac{\lambda}{2} \sum_{j=1}^d \theta_j^2}_{\text{Regularization term.}}$$

$\|\boldsymbol{\theta}\|_2^2$

- $\lambda$  is the regularization parameter ( $\lambda \geq 0$ ).
- No regularization for  $\theta_0$ .
- $\sum_{j=1}^d \theta_j^2$  =  $\|\boldsymbol{\theta}_{1:d}\|_2^2$  (magnitude of the parameter vector)
- $\sum_{j=1}^d \theta_j^2$  =  $\|\boldsymbol{\theta}_{1:d} - \mathbf{0}\|_2^2$  (pulling the elements toward 0)

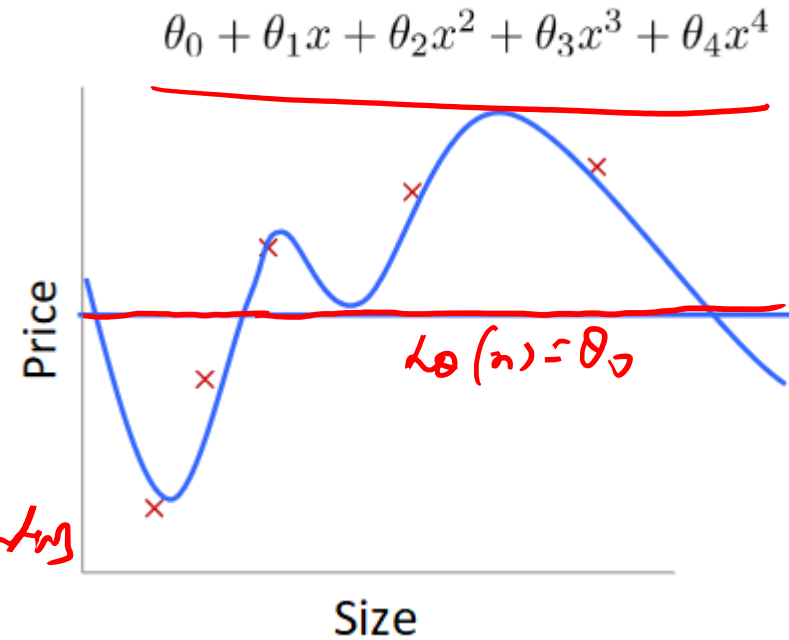
# Importance of Regularization Parameter

$$J(\theta) = \frac{1}{2n} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)})^2 + \frac{\lambda}{2} \sum_{j=1}^d \theta_j^2$$

- What if  $\lambda$  is set to an extremely large value?

$\theta_1 \approx 0, \theta_2 \approx 0, \theta_3 \approx 0, \theta_4 \approx 0$   
[we will end up penalizing the parameters  
very high].

$$h_{\theta}(x) = \theta_0$$



# Line Fitting

$$J(\theta) = \frac{1}{2n} \left[ \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)})^2 \right] + \frac{\lambda}{2n} \left[ \sum_{j=0}^d \theta_j^2 \right]$$
$$\frac{\partial}{\partial \theta} J(\vec{\theta}) = \frac{1}{n} \sum_{i=1}^n \left( \sum_{k=0}^d \theta_k x_k^{(i)} - y^{(i)} \right) x_j^{(i)} + \frac{\lambda}{n} \sum_{j=0}^d \theta_j$$

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- Fit by solving  $\min_{\theta} J(\theta)$

- Gradient descent with regularization:

- $\theta_0 := \theta_0 - \alpha \frac{1}{n} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_0^{(i)} \rightarrow 1$
- $\theta_j := \theta_j - \alpha \frac{1}{n} \sum_{i=1}^n (h_{\theta}(x^{(i)}) - y^{(i)}) x_j^{(i)} - \alpha \frac{1}{n} \lambda \theta_j$

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$$1 - \alpha \frac{\lambda}{n} < 1 \quad [\text{If } \alpha > 0, \lambda > 0, n > 0]$$

$$1 - \alpha \frac{\lambda}{n} \approx 0.99$$

So,  $\theta_j \times 0.99$  smaller than  $\theta_j$ .

Hence we are shrinking  $\theta_j$  at each iteration.

# Normal Equation

$$\theta = (X^T X + \lambda I)^{-1} X^T y$$

$$\min_{\theta} \{ \|y - X\theta\|_2^2 + \lambda \|\theta\|_2^2 \}$$

$$2 \left[ -X^T (y - X\theta) \right] + 2\lambda\theta = 0$$

$$-X^T y + X^T X \theta + \lambda\theta = 0$$

$$(X^T X + \lambda I) \theta = X^T y$$

$$\hat{\theta} = (X^T X + \lambda I)^{-1} X^T y.$$